



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE – (24AD3107)

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TITLE OF THE PROJECT	Medical Insurance Cost Predictor	
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Abstract

Medical Insurance Cost Predictor is a machine learning-based regression system designed to estimate individual medical insurance charges from demographic and health-related factors. The project uses the Medical Cost Personal Dataset, containing attributes such as age, sex, body mass index (BMI), number of children, smoking status, region, and insurance charges. The system applies data preprocessing and exploratory data analysis to understand distributions, relationships, correlations, and patterns associated with insurance costs. Categorical attributes are transformed into numerical representations before model development. The analysis examines the influence of factors such as smoking status, BMI, and age on insurance expenses, providing insights into characteristics associated with higher costs.

For cost estimation, the system applies Linear Regression, Random Forest Regression, and Gradient Boosting Regression. These models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score to compare predictive performance and identify the most suitable model for the dataset. The selected model is integrated into an interactive dashboard where users can enter individual attributes and obtain an estimated insurance cost. The dashboard also presents visual analytics, model performance comparisons, and feature-importance insights to improve interpretability. The primary objective is to demonstrate how machine learning and data analytics can be used to understand medical insurance cost patterns and provide data-driven cost estimates. The system is intended as an analytical tool for estimation and does not replace professional insurance or medical advice.

Signature of the Guide

HOD